PROPERTY (I) MODULES ARE K-INJECTIVE: THEY ADMIT NO MORPHISMS FROM ACYCLIC MODULES IN THE HOMOTOPY CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the filtration triangle with the representable-homological reduction to the quotients, the finite admissible filtration of each quotient with products of shifted duals as graded pieces, and the terminal Hom vanishing through lemma-hom-into-shift-dual-free and the exactness of the dual are all retained verbatim.

Editorial note. The label lemma-property-I-K-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The distinguished triangle of the property (I) filtration. Same chapter, Lemma lemma-property-I-sequence; Derived Categories, Lemma derived-lemma-representable-homological. Turns the short exact sequence of the filtration into a distinguished triangle and reduces the vanishing of Hom to the filtration quotients.
- Hom into shifted duals of the algebra. Same chapter, Lemma lemma-hom-into-shift-dual-free; the exactness of the duality functor. Computes the terminal case: Hom from an acyclic module into a product of shifts of A-dual vanishes because the dual is exact.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Prerequisite pointer invoked at proof line(s) 2637; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K4: sources/stacks-dga/pinned-original-source.tex, source lines 2366–2366. Prerequisite pointer invoked at proof line(s) 2650; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KR: sources/stacks-dga/pinned-original-source.tex, source lines 2598–2598. Prerequisite pointer invoked at proof line(s) 2635; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 014S: sources/stacks-dga/pinned-original-source.tex, source lines 1347–1347. Prerequisite pointer invoked at proof line(s) 2633; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module with property (I). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(N, I) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 014S). Let $F_\bullet$ be a filtration on I as in property (I). The short exact sequence of Lemma 09KR produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, I/F_iI) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules I/F_iI has a finite filtration by admissible monomorphisms, whose graded pieces are products of $A^\vee[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, A^\vee[k]) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 09K4 and the fact that $(-)^\vee$ is an exact functor. □